MIL-C-3191F
28 March 1984
SUPERSEDING
MIL-C-3191E
19 February 1970

MILITARY SPECIFICATION

CLOTH, FLANNEL, WOOL, LINING, 12 OUNCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of wool flannel lining cloth.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF AGRICULTURE (USDA)

CFR 7-Part 31 - Wool Standards

Methods for Determining Grade of Wool Top

(Application for copies should be adddressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20250.)

Laws and Regulations

Rules and Regulations Under the Wool Products Labeling Act

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials.

3.2.1 Wool. The wool shall be fleece or pulled wool, or any combination thereof not lower in grade than 60's U.S. Standard. The use of reprocessed wools, reused wools, or any wool waste or wool manufacturing by-products other than those resulting from the contractor's own current production of this cloth is prohibited. Garnets of any kind are prohibited.

3.2.2 Yarn. The wool shall be carded and spun on the woolen system into single yarns for both warp and filling (see 4.2.4).

3.3 Color. The color of the finished cloth shall be as specified (see 6.2 and 6.4) and shall be either stock or piece dyed to match the standard sample.

3.3.1 Mothproofing. Mothproofing of the wool material in accordance with MIL-C-43665 shall be accomplished during stock dying or piece dying as applicable (see 4.2.4).

3.3.2 Matching. The color and shade of the finished cloth shall match the standard sample (see 6.3) under artificial daylight having a color temperature of 7000 ± 500 K and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K.

3.3.3 Colorfastness. The finished cloth shall show fastness to perspiration, wet drycleaning and crocking equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to perspiration and wet drycleaning and show an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5. Testing shall be as specified in 4.2.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.2.4.

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TABLE I. Physical requirements

Width exclusive of selvage (minimum), inches	Weight per linear yard on a 56 inch width basis (minimum), ounces	Yarns per inch (minimum)		Breaking strength pounds, (minimum)		Weave
		Warp	Filling	Warp	Filling	
60	12.0	33	33	35	30	Four harness twill to the right - 2 up and 2 down

3.5 Finish. The cloth shall be scoured, fulled (carbonized if necessary), napped and sheared. The type and character of the finish shall be equal to that shown by the standard sample.

3.5.1 pH. The pH value of the water extract of the finished cloth shall be no less than 4.0 nor more than 8.0 when tested as specified in 4.2.4.

3.6 Shrinkage. The finished cloth shall shrink no more than 5.0 percent in the warp direction nor more than 3.0 percent in the filling direction when tested as specified in 4.2.4.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 50 yards. Each length shall be put-up on full width rolls as specified in 5.1.

3.8 Wool content. The finished cloth shall contain not less than 95 percent wool based on the dry weight of the specimen when tested as specified in 4.2.4.

3.9 Wool content label. The cloth shall be labeled in accordance with the Wool Products Labeling Act (see 2.1).

3.10 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than 1/2 inch in height (see 4.2.3.4). The marking shall be clearly legible in any indelible marking medium used commercially.

3.11 Workmanship. The finished cloth shall conform to the quality of product established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum acceptable point value.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Wool grade examination. The wool shall be visually examined for grade (see 3.2.1) in the scoured form, prior to dyeing (if dyed in stock form), picking or carding, by comparison with the applicable U.S. Standard (see 2.1). In the event of dispute resulting from this comparison, the wool grade shall be determined by a width method (wedge). The examination for grade shall be performed on a composite sample of 1 pound of scoured wool for each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot. The lot shall be unacceptable if any one sample does not conform to the requirement of 3.2.1.

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved wools are used (see 3.2.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.2.3 End item examination.

* 4.2.3.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section II of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.2.3.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.5), "D" for slubs and "C" for knots, shall be scored. The same yardage shall be given a

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through-lighting inspection from the back of the cloth towards a light source for holes and thin areas not previously scored on the face side of the cloth. Slight open places resulting from removal of burrs, shives or sisal or broken yarns not previously scored on the face side shall not be scored in through-lighting inspection. Defects on the selvage shall not be scored unless the body of the cloth is affected. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3,200 or less	8
3,201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 10.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 15.0 points. If one roll exceeds 15.0 points per 100 square yards, a second sample shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 15.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Degree and character of finish not equal to standard sample.
Width less than specified.

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4.2.3.2 Length examination.

4.2.3.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.3.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

* 4.2.3.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance (see 3.1 and 3.5). Any roll in the sample off shade or shaded side to side, side to center, or end to end, or any roll that does not have the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.2.3.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Back stamping missing from either or both ends.
Back stamping on wrong side.
Letters less than 1/2 inch in height.
Not labeled or ticketed in accordance with the Rules and Regulations Under the Wool Products Labeling Act.

4.2.4 End item testing. The cloth shall be tested for the applicable characteristics indicated in table II. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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*TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Yarn - single ply	3.2.2	Visual 1/
Mothproofing	3.3.1	2/
Colorfastness:		
Perspiration	3.3.3	5680
Wet drycleaning	3.3.3	5622
Crocking	3.3.3	5651
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Weave 1/	3.4	Visual
pH	3.5.1	2811
Shrinkage:	3.6	5558
Wool content (finished cloth)	3.8	2101

1/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

2/ As specified in MIL-C-43665.

* 4.2.5 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1132.

* 5. PACKAGING

* 5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1132.

* 5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1132.

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6. NOTES

6.1 Intended use. The cloth is intended for use in removable liners in men's and women's wool overcoats used by the Department of Defense.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Color required (see 3.3).
(c) Minimum length if other than specified (see 3.7).
(d) Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard samples, address the contracting activity issuing the invitation for bids.

6.4 Suggested dyestuff formulations. Suggested but not mandatory dyestuff formulations for Taupe 78 and Army Green 280 are as follows. The standard shade samples have been dyed with these formulations:

Taupe 78	Army Green 280
Acid Brown 19	Neutral Premetallized Dark Green 2BL
Acid Black 48, C.I. 65005	Neutral Premetallized Olive 3BL
Mordant Yellow 30, C.O. 18710	Neutral Premetallized Olive 2RL
	Acid Orange 86

* 6.5 Fabric defect scales. Fabric Defect Replica Kits are available from:

Sears Roebuck and Company
Department 817, (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Air Force - 99
DLA - CT

User activity:

Air Force - 82

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-3191F	Cloth, Flannel, Wool, Lining, 12 Ounce

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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